

PREAMBLE TO VOLUME 5

- a) The Tenderer's attention is drawn to the Instructions to Tenderers, Form of Tender, Standard Conditions of Contract, Particular Conditions of Contract, Specifications, and all other documents attached to this Contract, which shall be read in conjunction with the contents of Volume 5.
- b) The Works shall be delivered using BIM (as required in Particular Specification Appendix P2C Employer Information Requirements (EIR) and General Specification G1.2 Information Management) as will the wider TWRP Project.
- c) The primary means of exchange of engineering (and other) information for the project between the S.O. and the Contractor (and vice versa) shall be through the exchange of digital information rather than conventional 2D drawings and isolated, disconnected data.
- d) The schematic design of the Works, including P&IDS, electrical single line drawings and the like are by necessity to be presented in 2D.
- e) The Contractor's attention is drawn to the Particular Conditions of Contract, Clause 1.1 (n) regarding the meaning of the term Drawings.
- f) The Drawings prepared by the S.O. describe the design intent at the time of tender. The Contractor's responsibility is to develop the BIM tender model fully to support the post-tender delivery of the Works as specified in the Particular Specification Appendix P2C EIR and General Specification G1.2 Information Management.
- g) The civil and structural (C&S), MEICA (Mechanical, Electrical, Instrumentation, Control & Automation) and architectural designs included in the Drawings provide a reference to allow the Contractor to appreciate the required operability, maintainability and general arrangement requirements of the plant and equipment. The Contractor shall develop the design for construction in compliance with the Specifications, subject to the approval of the S.O..
- h) Included in Volume 5 is a consolidated BIM Model and pdf files produced from the native BIM model files. These are:
 - 1) Volume 5, Part IA - 2D Drawings:
 - .pdf format files of 2D drawings – extractions of model details, general notes, schematics, legend sheets and schedules for the Works.
 - 2) Volume 5, Part IB – Standard Details:
 - .pdf format files of standard engineering details for incorporation into the detailed design by the Contractor.
 - 3) Volume 5, Part II – BIM Model:
 - The “.i.dgn” format model files describing the design intent for the Works (readable with Bentley view (Bentley View CONNECT can be downloaded in the link www.Bentley.com)).
 - The “.iModel” is created in v2.0 and can be opened by Navigator Connect Edition v10.02.
- i) The Tenderer shall follow the S.O. model authoring software(s) as specified in the Particular Specification Appendix P2C EIR.
- j) The Tenderer shall extract the information necessary from the BIM model and 2D drawings for the preparation of his Tender proposal. No other drawings will be provided by the Board at the time of Tender.

- k) After the Commencement Date, the S.O. will provide native files to the Contractor, who shall be responsible for developing the content of these to meet the requirements of the Contract and for the approval of the S.O..
- l) The Contractor shall note the provision of additional Models in Volume 6 for reference. These additional models include indicative C4B scope of work developed by the S.O to appreciate the required operability, maintainability and general arrangement requirements of the plant and equipment. These additional models include space reserve for future equipment. The Contractor shall refer to these space reserves and shall allow for them in the Contractor's design and layout.
- m) The Contractor shall note the access routes (also known as process mechanical service space) in the Model and shall allow for a clear, unrestricted pathway in the Contractor's design and layout.
- n) The successful Contractor will be provided with both the workspace and datasets used by the S.O. to create the Tender models. The Contractor shall modify the workspace and expand the datasets as required to create the virtual construction and As-Built BIM models upon S.O's approval.
- o) Once complete, the BIM Models shall be classified as 'As-Built' BIM Models and provided to the S.O. as described in Particular Specification Appendix P2C EIR and General Specification G1.2 Information Management.
- p) The following sections provide more information regarding Volume 5 Part II and, where applicable, the underlying structures to aid understanding and interpretation.
- q) **Model Files**
 - 1) The model files listed in Table 1-1 detail the infrastructure, plant and equipment described in the BIM Tender model.
 - 2) Abbreviations:
 - ARC – Architecture
 - STR – Structural
 - BS – Building Services
 - CIV - Civil
 - ELEC – Electrical
 - ICA – Instrumentation & Control Automation
 - MECH – Mechanical

Table 1-1 Model Files Included in Volume 5

Item No.	Model File Name	Description	Comments
1	131101-WSW41-M3-Z-8690	C4B Overall BIM Tender Model (Includes multidisciplinary information regarding the facilities)	This model file combines all other model files.
<i>Item No. 1 includes the following Models</i>			
1.1	131101-WSW41-M3-Z-8900	C4B Sitewide Pipe Racks	
1.2	131101-WSW41-M3-C-8902	SSN Minor Drains	
1.3	131101-WSD11-M2-S-8901	C4B Working Area (THP)	
1.4	131101-WST02-M2-S-8901	C4B Working Area (SSN)	

Table 1-2 Saved Views included in the Model Files

Saved View Name	Description	View
00_<FacilityCode>_<Discipline>_HomeView	Front Page with Date Stamp	2D view
01_<FacilityCode>_<Discipline>_Isometric	Overall Isometric View	3D View
02_<FacilityCode>_<Discipline>_FP	Floor Plan View	3D View
03_<FacilityCode>_<Discipline>_SE	Section view	3D View

Architectural discipline to have all the saved views in Table 1-2 and the rest to have the first two saved views as the minimal.

Below are the procedures to navigate to saved views

- In 'Manager View Groups" (Figure 1) select "Home View" (Figure 2) to see the Date Stamp.

Figure 1

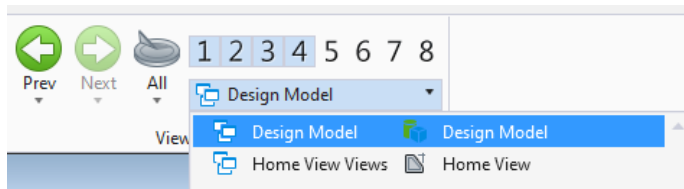


Figure 2

TUAS WATER RECLAMATION PLANT (TWRP) DTSS PHASE 2	
MODEL FILE NAME	131101-WSW41-M3-Z-8690
MODEL DESCRIPTION	OVERALL BIM TENDER MODEL
CONTRACT NO	C4B
CONTRACT DESCRIPTION	DIGESTION PT AND SSN REMOVAL FACILITIES
MODEL ISSUE DATE	20/05/2022
MODEL REVISION	T1
MODEL STATUS	TENDER
MODEL PURPOSE	FOR INFORMATION
NOTE	NOTE

To See all the rest of the view

- Under Primary group under Home ribbon, Select "Level Display" (Figure 3) and navigate to "Level Display" (Figure 4) window and select any of the model (Figure 4).

Figure 3

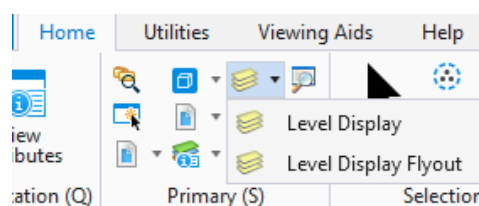


Figure 4

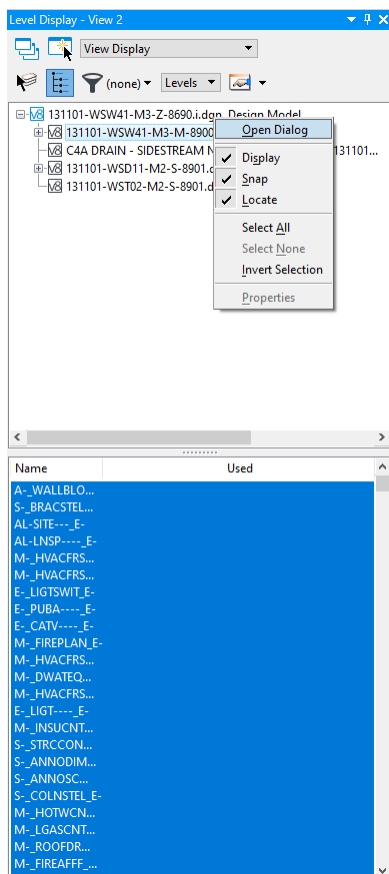


To turn on and off the various linked model:

Select the design model in Level Display window, expand by clicking on the “+” sign to see all the references (Figure 5).

To turn on and off the links (references) / Levels, toggle (Figure 5) by selecting the square box as highlighted using mouse right button.

Figure 5

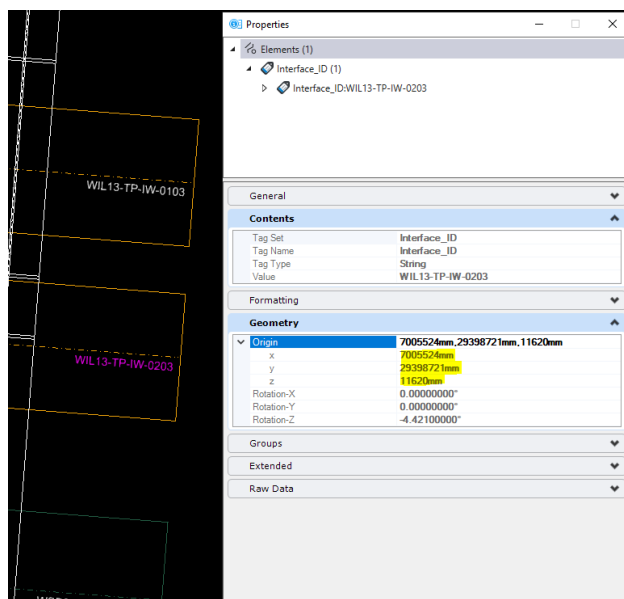


r) Tie-in / Interface Points

Interface points can be read from the BIM Tender Model by clicking on an interface point tag. Volume 6 Reference Models can also be read in conjunction to identify the location of interface points between C4B and Other contractors. These can sometimes be hidden so the view may need to be changed to wireframe to view, and by reading the X, Y and Z coordinates as per Figure 6 from the element information tab.

Interface points are identified as shown in Figure 6 and tagged at all points where a service has crossed over into another contract.

Figure 6



- The Tie-in register included in Particular Specification P2 Preliminaries and General, details the contract Interfaces. The Contractor shall update the register and associated attributes in the BIM Model to the satisfaction of the S.O. as detailed design progresses.

Element Information (Attributes)

- s) In each BIM Tender model, the element information tab provides a limited number of attributes for any selected item. For example, in the "Identification" section the equipment tag number can be found under the Tag Information section or TUAS identification section.
- t) The attribute information provided in the BIM tender Model reflects the information available at the time of Tender, which is not dependent on the Contractor's procurement activities and is intended to convey the basic design criteria to assist the Tenderer in understanding each element.
- u) The element whichever is for space proof is marked as "Not part of C4B Contract – For information Only" on the "TUAS | Tag Number" attributes in the model with 60% element transparency.
- v) Pipework less than 80mm Nominal Bore is not shown in the model.
- w) Attributes, which are left blank in the BIM Tender Model or are provided with 0 or 1 or which are auto generated by the model, shall be provided by the Contractor after design or procurement of specific products.
- x) The Contractor shall provide, as a minimum, all attributes in Figure 7 as part of compliance with the requirements given in Particular Specification Appendix P2C EIR and General Specification G1.2 Information Management.

Figure 7

General		Geometry		Material		Extended	
Description	PIPE SPIGOT SOCKET 1600 DIPU WSW00-061-NW-0009-1600-DIPU	Origin	7047208 66.29405476 66.29405476	Assigned Materials	(None)	Model	Default
Cell Name		x	7047208.66			Last Modified	25-Apr-18 2:01 PM
Cell Type	Graphic	y	29405476.66			Modified	Modified
Class	Primary	z	-1551.08			New	Not New
Number of elements	2	Rotation-X	.0571°			Locked	Unlocked
Template	None	Rotation-Y	-.0044°			Display Style	☐ From View
Annotation Purpose	False	Rotation-Z	85.579°				
		Scale X	1.00000				
		Scale Y	1.00000				
		Scale Z	1.00000				

Tag Information		Design Values		Record		Operating Conditions	
Name	PIPE SPIGOT SOCKET 1600 DIPU WSW00-061-NW-0009-1600-DIPU	Design State	New	Modification Time	25-Apr-18 8:01 AM	Fabrication Category	7928
Project Prefix	WSW00	Insulation Thickness	0 mm	Modified By	DW046264	Length	
Unit Name	061	Component State	New	Create Time	26-Mar-18 1:51 PM	Shop Field	
Material		Design Length Center T	0	Created By	DW046264	Code	
		Design Length Center T	0			Weight	0 kg
		Design Length Center T	-265			Grade	0
		Nominal Diameter	1600 mm			Inside Diameter	0 mm
		Nominal Diameter Run	1600 mm			Wall Thickness	0 mm
		Rating				Size	
		Schedule				Length Effective	7663
		Specification	DIPU			Sequence Number	
		Standard	BS EN 545			Pipe Flange Type	
		CONSTR_STATUS					
		4D Sequence Code	WSW00.061.NW..01				
		P6 Sequence	01				
		QTO	14.1				
		SD QTO Code	WSW00.061.NW..14.1				

Design Conditions		General Info		Miscellaneous		Raw Data	
Model	DW046264	Paint Code	DI SPIGOT/SOCKET PIPE	Notes		Element ID	306660
Designer		Description	DI SPIGOT/SOCKET PIPE	Unit Number		Size	239
Nominal Size		Elevation	0 m	Catalog Name		File Position	4008794
		Dry Weight	0 kg	Element Geometry		Linkages	4
		Manufacturer		EC Class Name	PIPE	XAttributes	5
		Material	DIPU	Branch Code			
		Order Number		Transformation Matrix			
		Stock Number		Component Name	PIPE SPIGOT SOCKET 1600 DIPU WSW00-061-NW-0009-1600-DIPU		
		Total Weight	0 kg	Line Number	WSW00-061-NW-0009-1600-DIPU		
		Created Timestamp	26-Mar-18 1:51 PM	Spool Id			
				Origin Code	1		

y) C4B Working Areas

- 1) C4B Working Areas for various facilities are indicated in the BIM Tender Model. The elements are marked as "C4B WORKING AREA" on the General attributes.
- 2) For the Digestion Pre-treatment Facility, the main C4B Working Area at the THP Courtyard is shown in the model. For the location of other equipment (e.g. feed pumps, compressed air system, electrical and ICA panels, and etc.) within the Biosolids Buildings, the indicative position can be viewed in the Volume 6 Reference Model. The final location of equipment, pipe works and cabling works within the Biosolids Buildings shall be coordinated with Other contractors.
- 3) The main C4B Working Area for Sidestream Nitrogen Removal Facility is shown in the BIM Tender Model. The facility shall be design and build within the working area. For works to be carried out outside the working area (e.g. connection to interfaces point), the Contractor shall coordinate with Other contractors to design and build the connections to the Sidestream Nitrogen Removal Facility.

System Requirement

For the System Configuration requirement, refer to www.bentley.com or

https://communities.bentley.com/products/projectwise/project_review/w/wiki/23572/6-technical-requirements

END OF SECTION